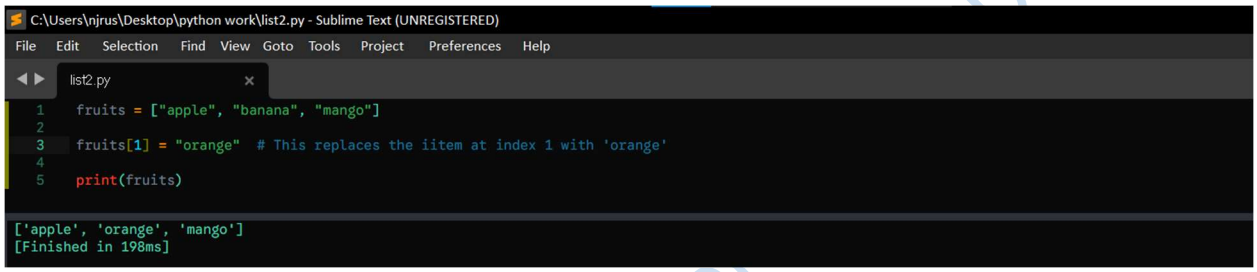
Working With Lists

In the previous lesson, we looked at two of important skill you need to build confidence with lists. We looked at how **you can access items in a list** and **how indexing work in lists**. We will now proceed by looking at two more skills:

- Changing Items in a List.
- Adding & Removing Items to & from Lists.

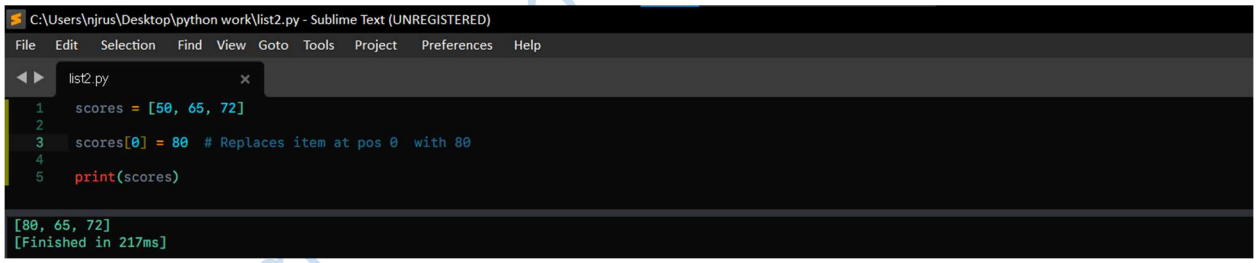
1. Changing an Item in a List

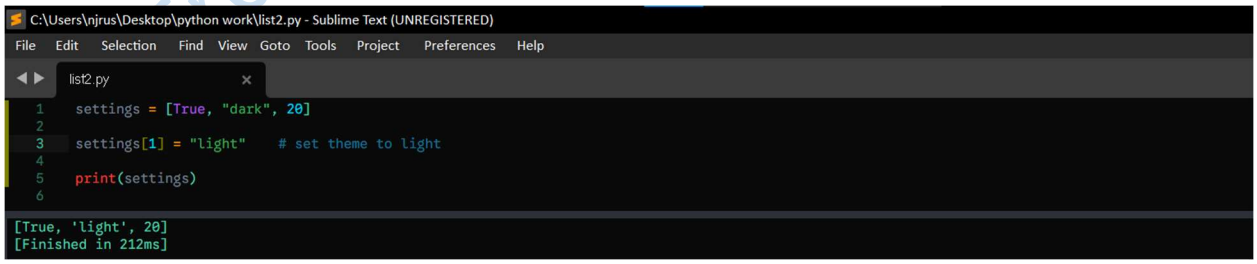
Lists are **mutable.** This means their contents can be changed after the list has been created unlike **tuples**. To change an item, we use its index and assign a new value to that position.

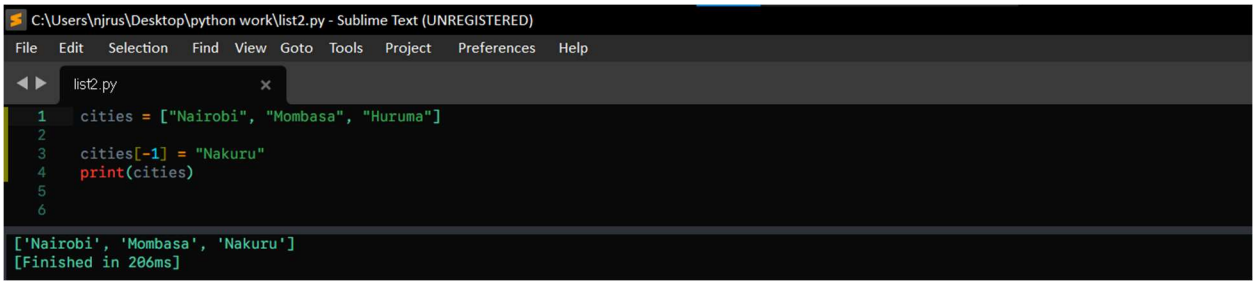


The item at index 1 was originally **"banana"**. The assignment `fruits[1] = "orange"` replaces that value with **"orange"**.

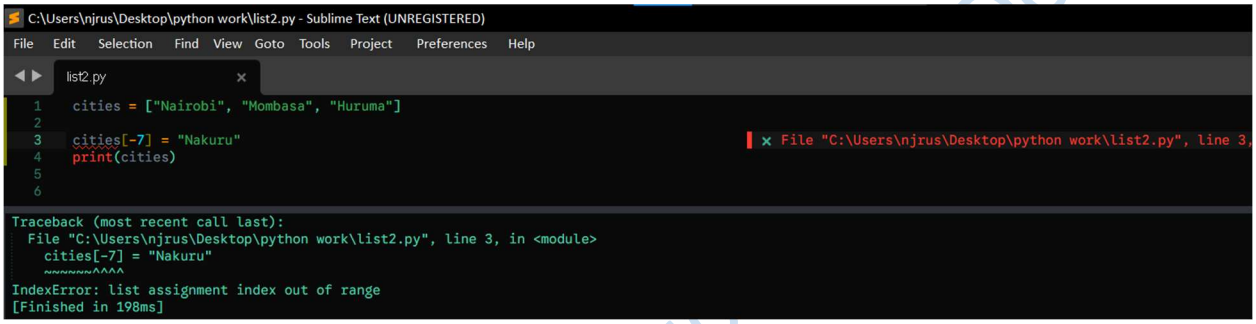
See some more examples below involving changing items:







Note: Changing items in a list does not create a new list. It only changes the item in that position in the existing list. In addition, trying to change an item using an index that does not exist will cause an error. See below:



To avoid such errors, use indexes that map to valid positions in the particular list you are trying to change items from.

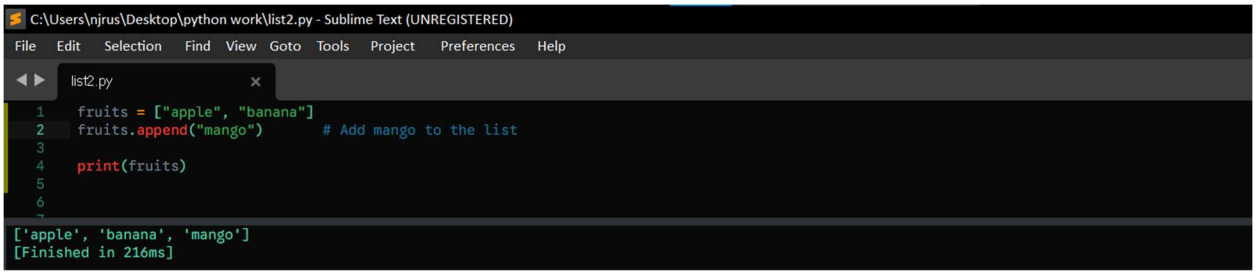
2 (a). Adding Items to a List

Because lists are **changeable,** we can also add new items after a list has been defined. Python provides several methods for doing this. The most common are:

- append()
- insert()
- extend()

The append() Method

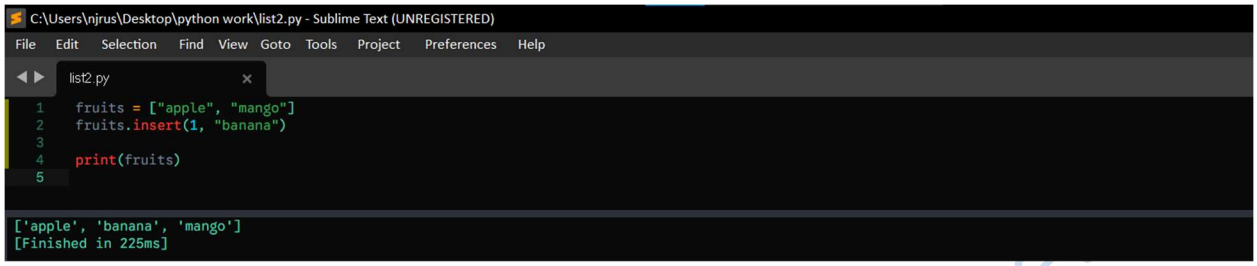
The **append() method** is used to **add one item to the end of a list.** It is one of the most commonly used list methods.

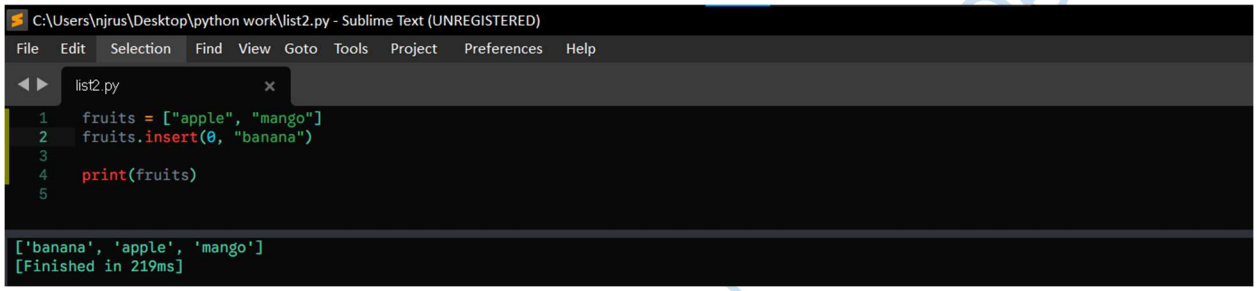


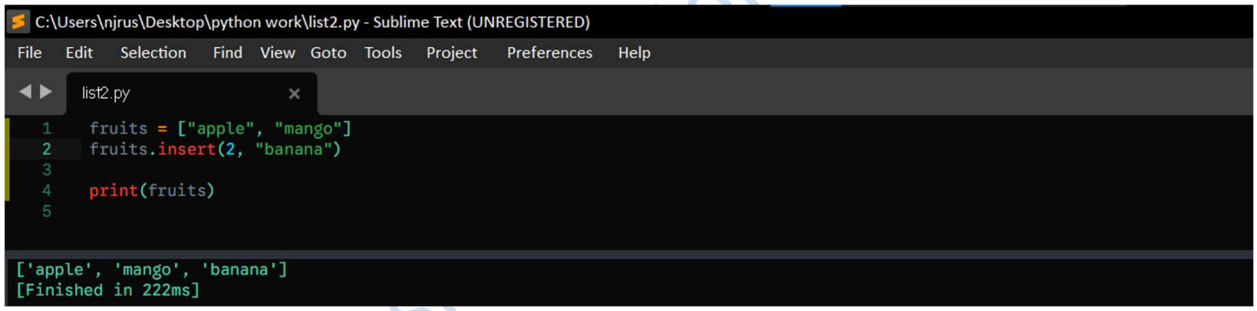
Note: The append method adds an item to the end of the list

The insert() Method

The **insert() method** is used to **add one item at a specific position**. It takes two arguments: the **index where the new item should be placed** and the **value to add.**

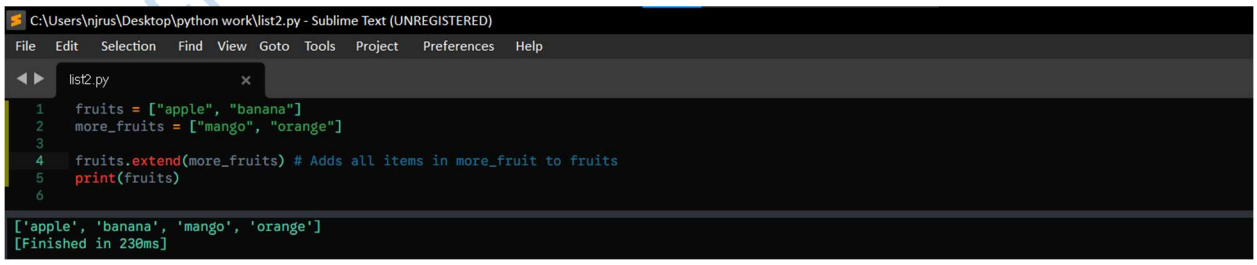






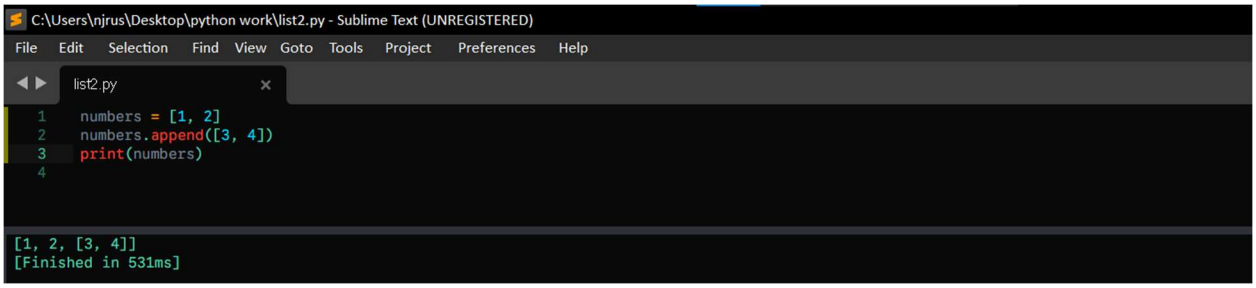
The extend() Method

The **extend() method** is used to **add all the items from another list, to the end of the current list**. It is useful when you want to combine several values with an existing list.

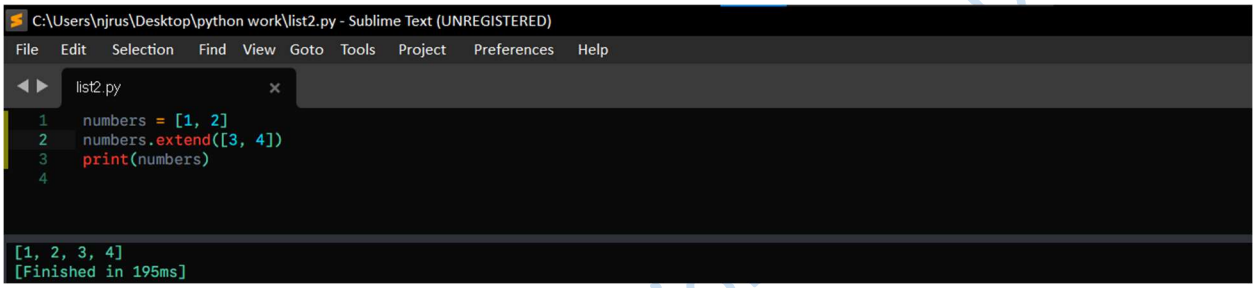


The append() and extend() methods are easy to confuse because both add values to the end of a list. To avoid this, keep in mind that whatever you pass to append() as an argument, append treats it as

single item and is added inside the original list.



On the other hand, extend() add all items available in the list you pass as an argument to the existing list.



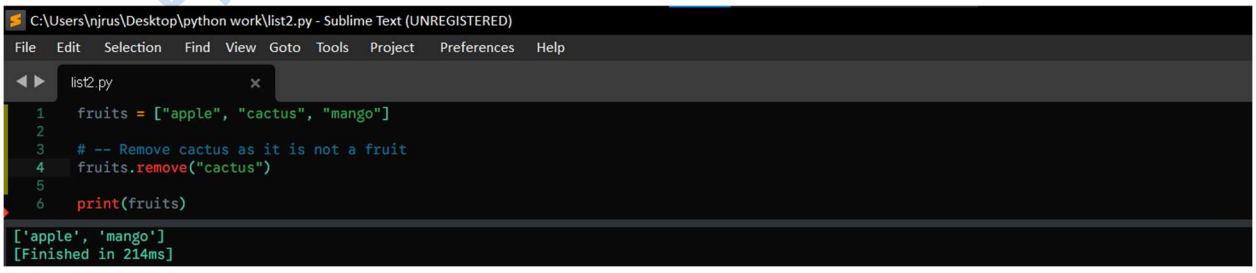
2 (b). Removing Items from a List

You can also remove items from a list! Python provides several ways to do this. The best choice depends on whether you know **the value you want to remove**, the **position of the item**, or whether you want to **remove everything**.

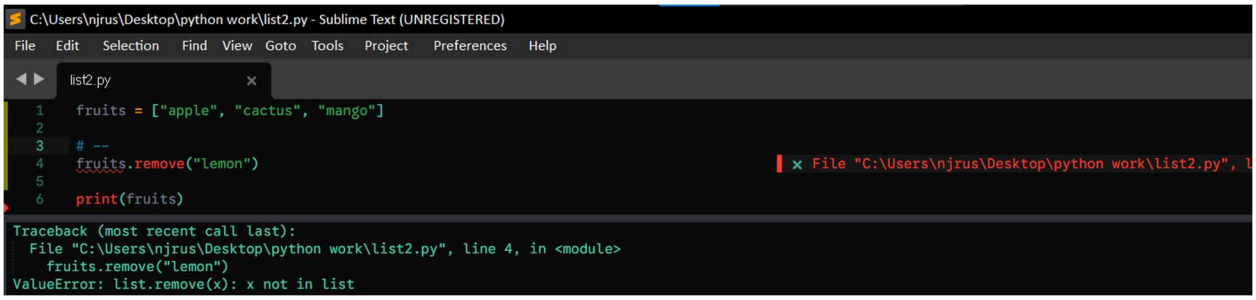
The remove() Method

The **remove() method** is used to remove an item from a list by value. That is, if you know item you wish to remove, you can pass it as an argument to the remove method. This removes the item from your list.

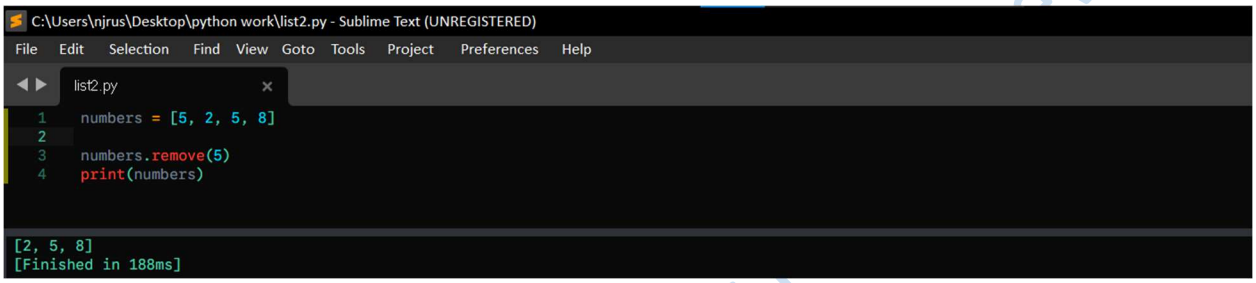
See below:



Note that if the value you pass to remove() does not exist in the list, an exception will occur.



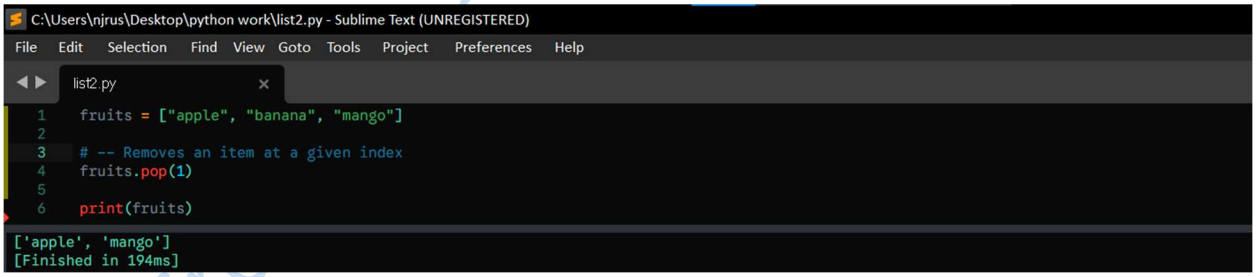
Since lists can contain duplicates, what happens when you try to remove an item that appears more than once?



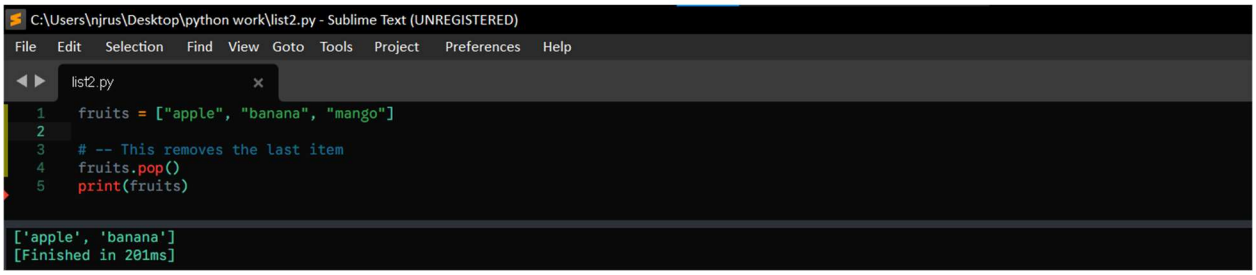
For duplicate items, remove() removes **only the first** matching occurrence.

The pop() Method

The **pop() method** is used to remove an item from a list by index. That is, given an index, pop(index) removes the item at that index in the list.



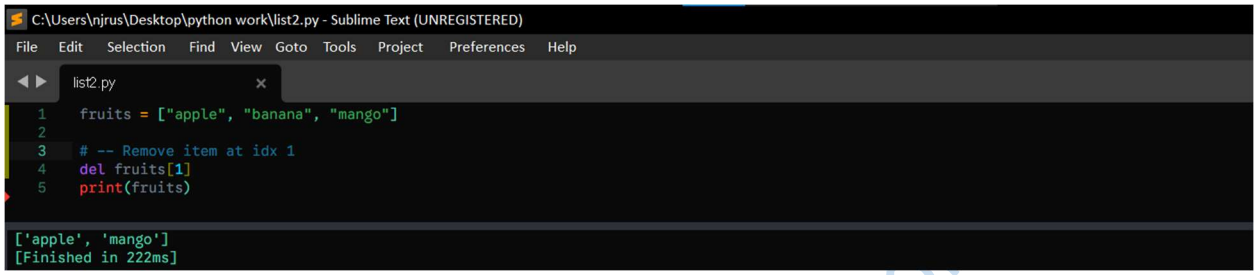
The pop() method can also work without an index. However, this removes the last item in that list.



If you pass an index that does not exist to pop(), an IndexError will occur. Use valid indexes to avoid such errors.

The del Statement

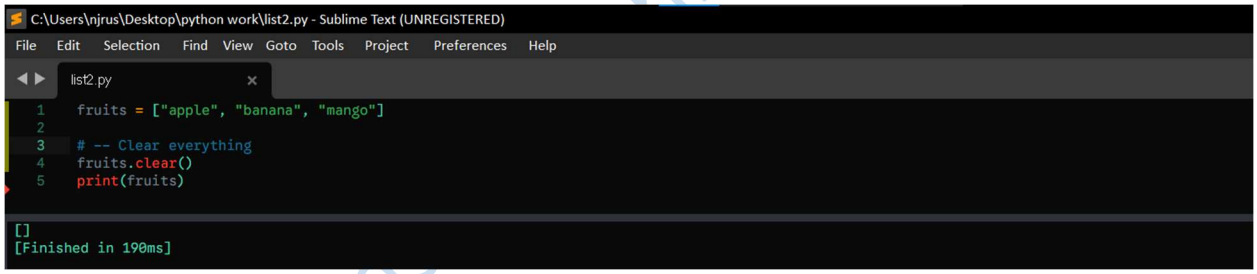
Just like the pop, **del** is used to remove an item at a specific index.



In addition, using an index that does not exist will cause errors.

The clear() Method

The **clear() method** us used to remove every item from a list.



Try Yourself

Instructions

- **Work through the instructions in the order given.**
- **Do not reset the list between steps. Each step must be performed on the list produced by the previous step.**
- **Write the Python statement(s) you would use for each instruction.**
- **Keep track of how the list changes after every step.**

Question — Music Playlist

You are managing a music playlist. Start with the following list:

```
songs = ["Dreams", "Tattoo", "Spirit", "Solo", "Bye", "Alone"]
```

Perform the following operations in the exact order shown. Each operation changes the list used by the next operation.

1. Change "Spirit" to "Through The Static".

2. Add "Glimpse of Us" to the end of the list.

3. Add "Faded" to the beginning of the list.

4. Remove "Solo" from the list.

5. Change the last item in the list to "On My Way".

6. Remove the second item in the list using its index.

7. Insert "The Nights" between "Through The Static" and "Bye".

8. Remove the first item in the list using its index.

Final List:

```
["Tattoo", "Through The Static", "The Nights", "Bye", "Alone", "On My Way"]
```

From Basics to Junior Python Dev